

PRIVATE EQUITY

TRANSACTION MODEL

PYTHON RISK

TAKE-PRIVATE REVERSE LBO & PROBABILISTIC UNDERWRITING

Project Redwood | Fictional Private Equity Case Study

Core underwriting question

What is the highest price a sponsor can pay while preserving return hurdles, debt capacity and downside protection?

Investment committee snapshot

Seller ask	8.5x EBITDA
Base-case return	22.3% gross IRR
Deterministic max bid	8.1x EBITDA
Risk-adjusted bid	8.0x @ 50% probability
Probability at ask	32.9% hit 25% IRR

8.5x

Seller ask

Entry EV / EBITDA

22.3%

Base-case gross IRR

vs 25.0% hurdle

8.1x

Deterministic max bid

IRR hurdle is binding

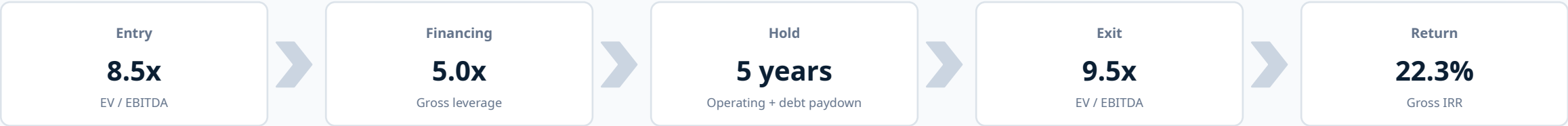
10,000

Synthetic deal paths

Correlated underwriting scenarios

The seller ask is a pricing question, not just an LBO output

The model starts with transaction mechanics, then asks whether the capital structure and exit case support the required sponsor return.



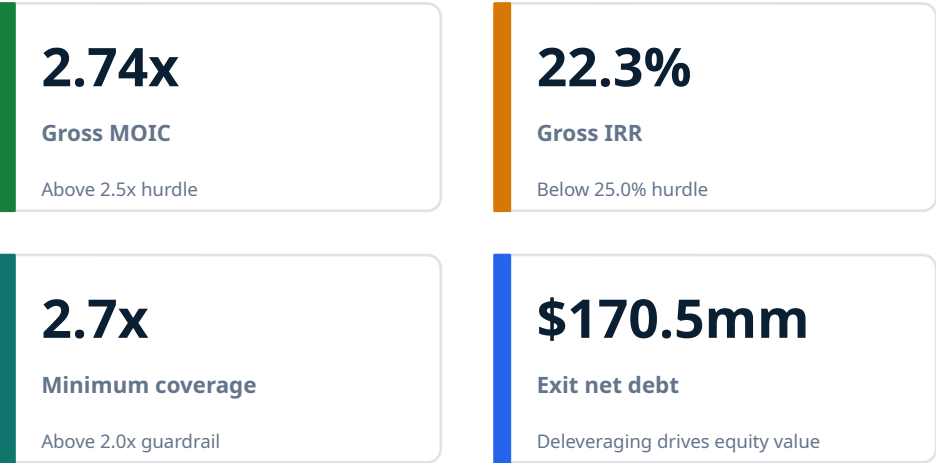
Sources & uses at close

Uses (\$mm)	Amount	Sources (\$mm)	Amount
Equity purchase price	479.9	Term Loan B	247.0
Refinance net debt	45.0	Second-lien	61.8
Minimum cash + fees	26.7	Management rollover	19.4
Total uses	551.5	Sponsor equity	223.4

Funding mix



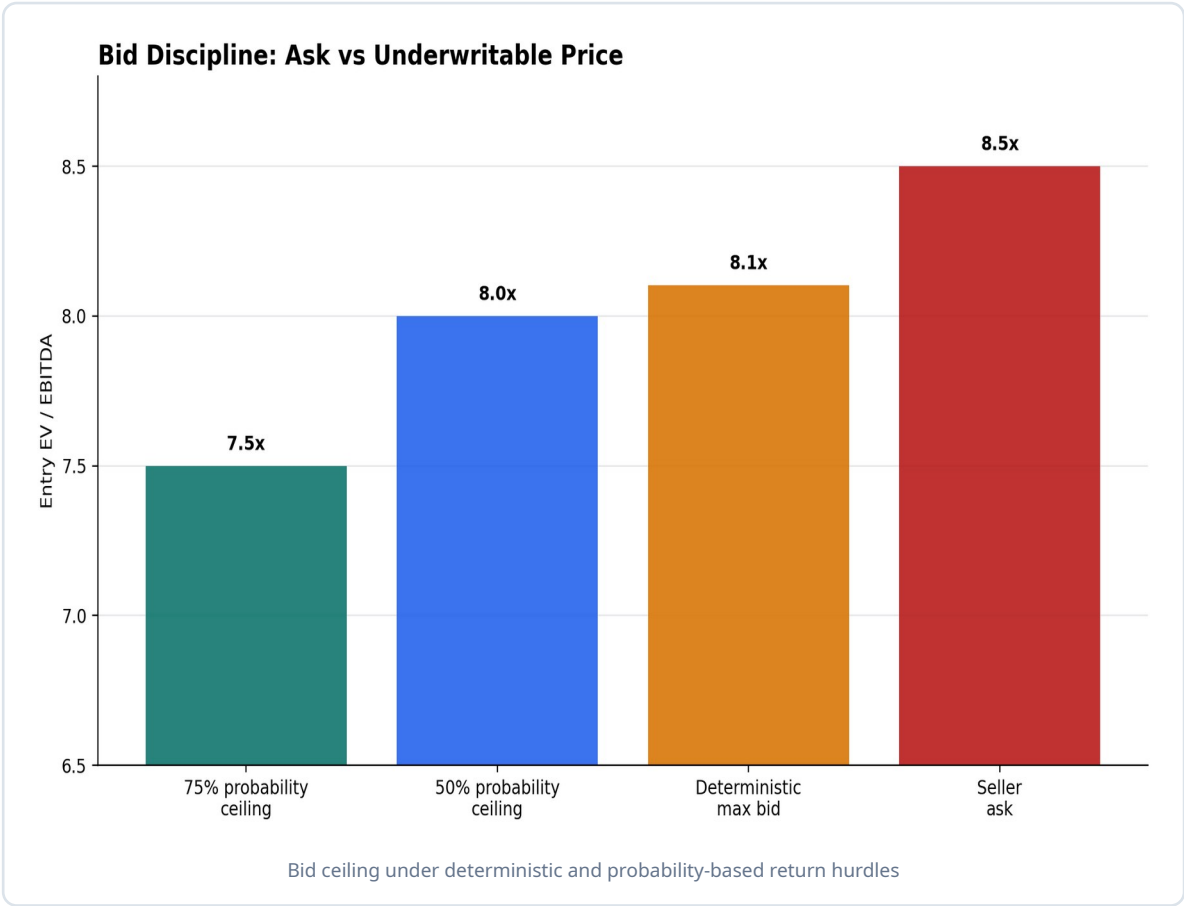
Deterministic underwriting readout



Key point: the base case is investable-looking, but the asking price does not clear the sponsor IRR hurdle. That creates a real bid / negotiate / walk-away decision.

Reverse LBO: solve the price before making the bid

Instead of forcing 8.5x to work, the model backs into the highest price consistent with sponsor return hurdles.



The bid gap

8.1x

vs

8.5x

maximum deterministic entry multiple seller ask

Implied bid gap

(0.4x) / (\$24.5mm EV)

If the sponsor still pays 8.5x, one of these has to improve to reach 25% IRR:

+9.1%

Exit EBITDA

10.4x

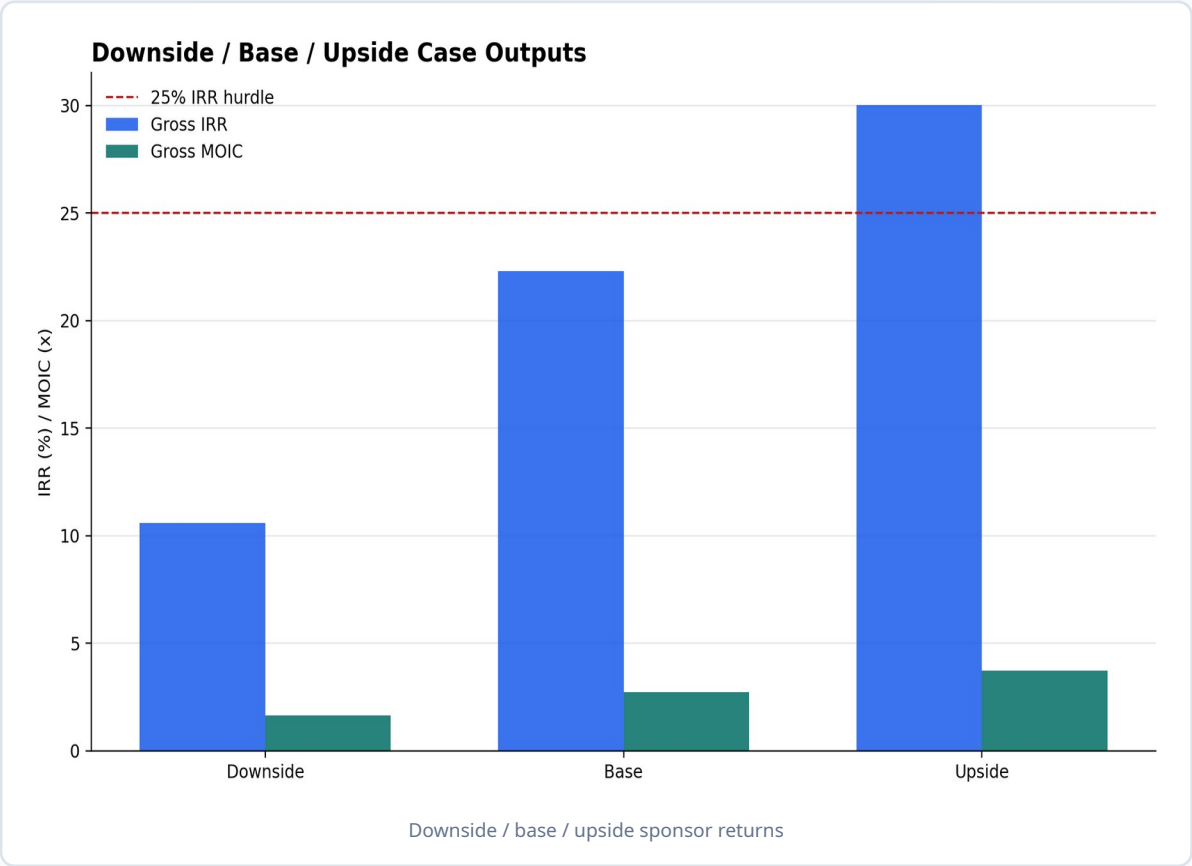
Exit multiple

\$76.3mm

Additional debt paydown

Downside protection matters as much as headline returns

The deal remains serviceable in the deterministic downside case, but returns compress sharply when growth, margins and exit valuation weaken together.



Scenario scorecard

Scenario	IRR	MOIC	Min coverage
Downside	10.6%	1.65x	2.16x
Base	22.3%	2.74x	2.71x
Upside	30.0%	3.72x	3.05x

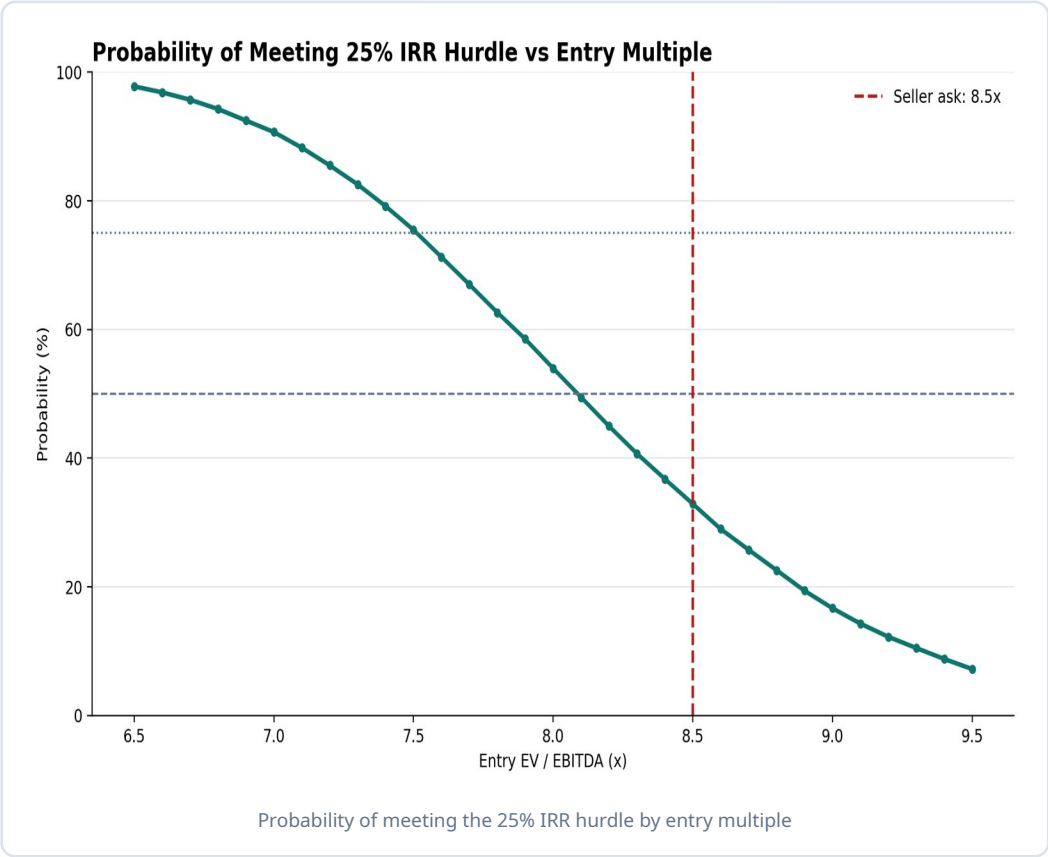
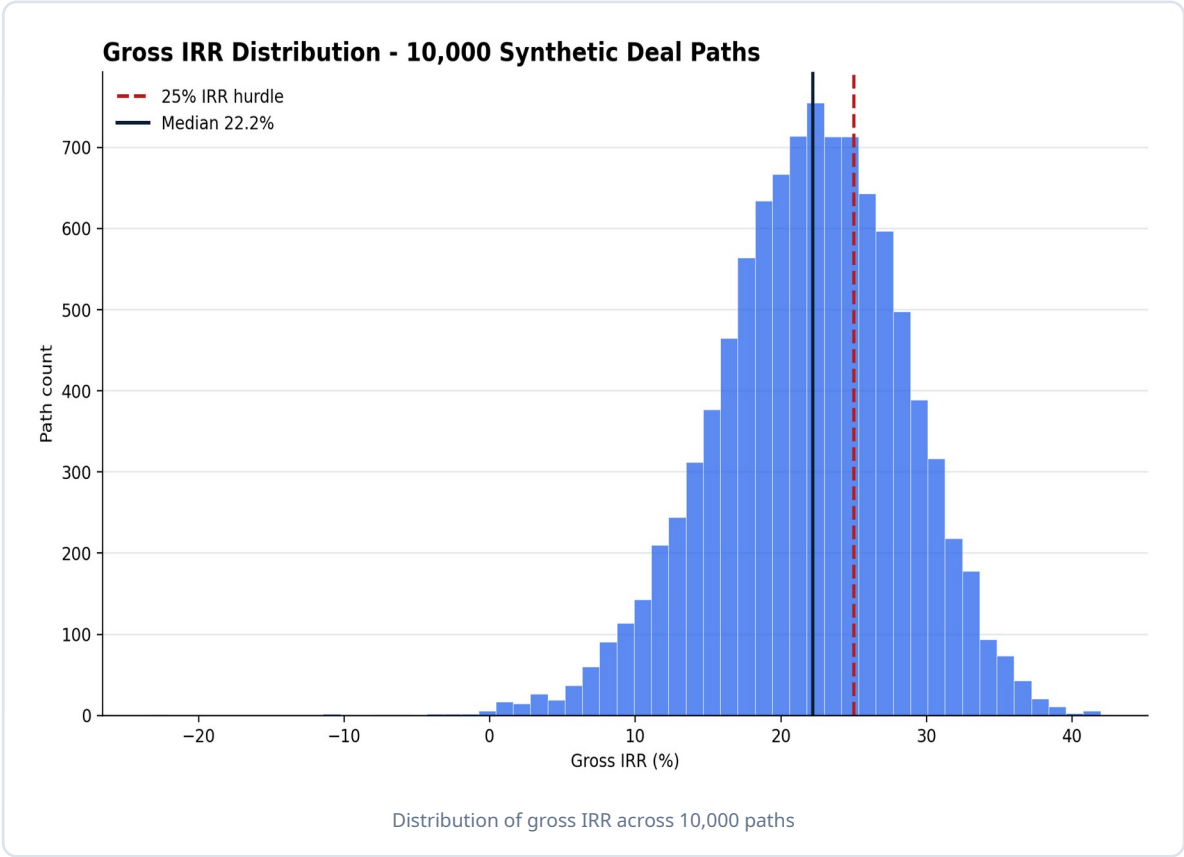
Credit guardrails in the 10,000-path simulation



Interpretation: the capital structure is not the main reason to reject the ask. The bigger issue is price relative to the required equity return.

Probabilistic underwriting: one base case is not enough

10,000 correlated synthetic paths vary growth, margins, rates and exit valuation, then pass every outcome through the same LBO debt waterfall.



32.9%

At 8.5x seller ask

of paths clear 25% IRR

8.0x

50% probability ceiling

Risk-adjusted bid

7.5x

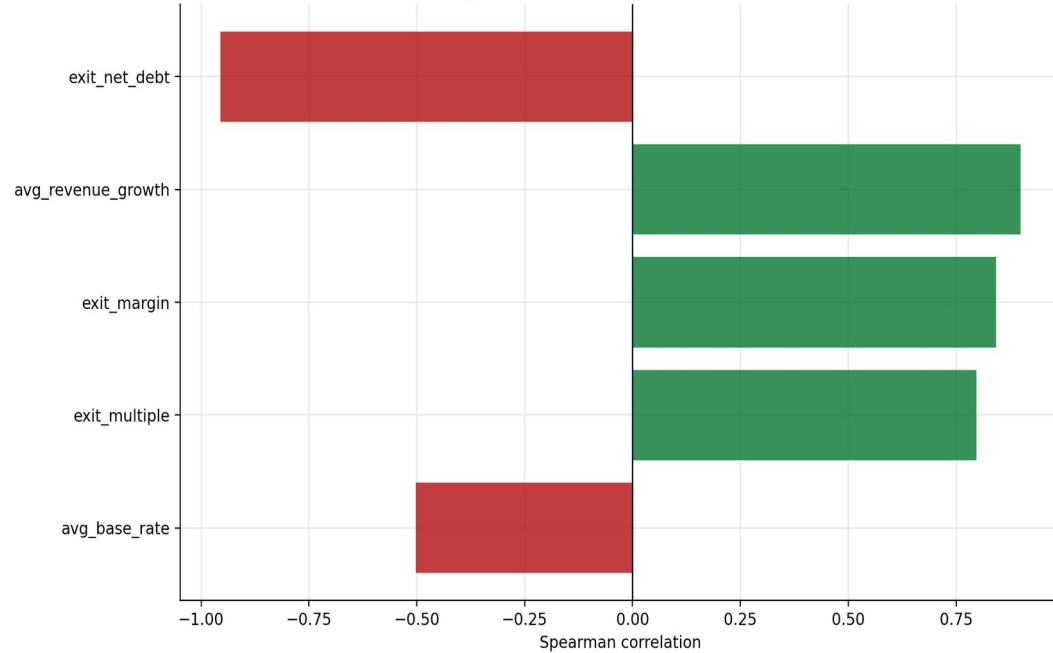
75% probability ceiling

Higher confidence = lower bid

Investment committee conclusion: bid discipline, not model engineering

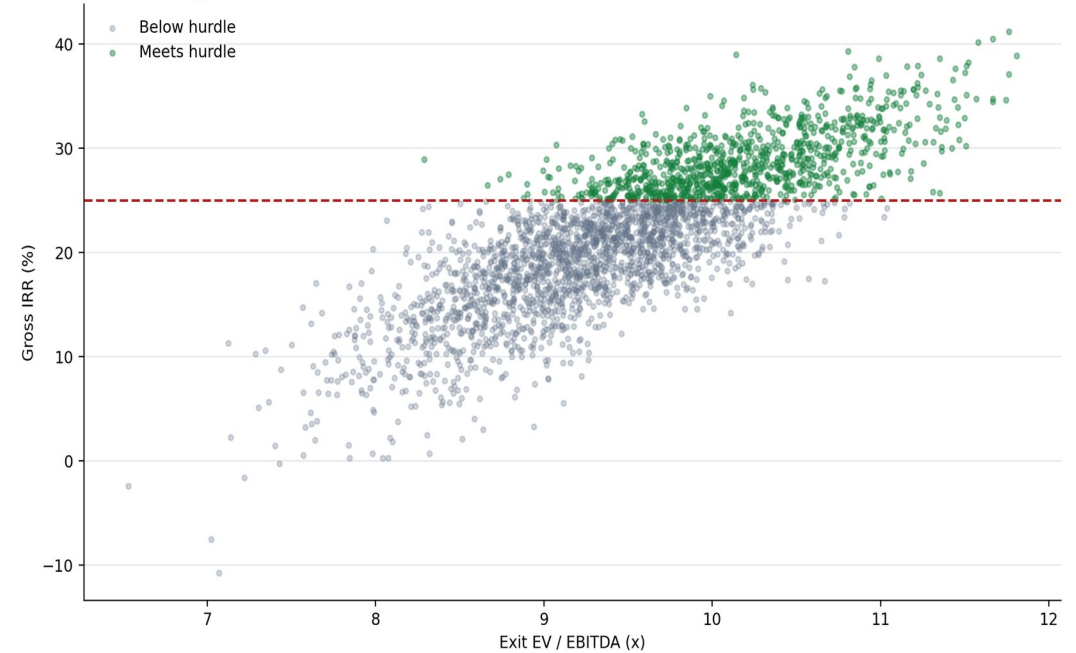
The framework connects valuation, leverage, operating execution and probability of success into one decision.

Deal Outcome Sensitivity - Rank Correlation with Gross IRR



Spearman rank correlation with gross IRR

Exit Multiple vs Gross IRR



Exit multiple vs realised gross IRR

IC takeaway

The 8.5x ask is possible, but not the most defensible price under the selected 25% IRR hurdle. The model supports ~8.1x deterministically, ~8.0x at a 50% probability threshold, and ~7.5x at 75%.

Deliverables: Excel LBO + Reverse LBO | Python risk engine | 15 automated tests | interactive HTML dashboard | reproducible CSV outputs